AWS VPC PEERING

NETWORKING LAB

PROJECT REPORT

Configuration
Validation
Findings

Secure private communication between two Amazon Virtual Private Clouds using VPC Peering.

This report records the network design, route-table configuration, security controls, connectivity tests, and controlled failure/recovery exercise completed during the AWS lab.

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Project focus	VPC Peering and private EC2 connectivity	**Environment**	Amazon Web Services (AWS)

PROJECT AT A GLANCE

2 VPCs Separate private networks	**1 active peering** Private inter-VPC path	**Two-way ping** Successful in both directions
Bastion SSH App-B access validated	**Security test** Failure and recovery observed	**Private instances** App-A and App-B

Submission note: verify all screenshots, review the final resource state, and complete the cleanup checklist before submitting.

Contents & Project Dashboard

Use the links below to jump to each report section in a PDF reader that supports internal links.

1. Executive summary	2. Architecture and resource inventory	3. Routing and peering configuration	4. Security group design
5. Validation results	6. Screenshot evidence	7. Cleanup checklist	8. Learning outcomes

Validation snapshot

PEERING	FORWARD PING	REVERSE PING	BASTION SSH
ACTIVE	PASS · 0% loss	PASS · 0% loss	PASS

What the lab demonstrates

VPC Peering enables private IP communication between two VPCs when the connection is active, both route tables include the peer CIDR, and security groups permit the required traffic. The controlled ICMP test demonstrated that a network path alone does not guarantee reachability: security rules still determine which traffic is allowed.

Evidence included in this report

The screenshot section contains available lab captures. Compare each image against its caption and replace any unclear photographic capture with a direct screenshot from the AWS console or terminal.

REPORT NAVIGATION TIP

Click a section name in the contents grid to jump to that section. Use your PDF reader’s page thumbnails for quick navigation.

1. Executive Summary

The objective was to connect two isolated Amazon VPC networks using VPC Peering, configure routes and security group permissions, and validate private communication between EC2 instances. The peering connection became active, ICMP tests succeeded in both directions, and SSH access to App-B through the bastion host was verified.

Key outcomes

• Two VPCs used non-overlapping CIDR ranges.

• Private route tables routed the peer CIDR through the active peering connection.

• App-A and App-B exchanged ping packets in both directions.

• SSH access to App-B through the bastion succeeded.

• Removing the App-B inbound ICMP rule caused ping to fail; restoring it recovered connectivity.

The report reflects results confirmed during the lab. Confirm resource state and complete cleanup before final submission.

2. Architecture and Resource Inventory

VPC A contains the public bastion subnet and App-A private subnet. VPC B contains App-B in a private subnet. VPC Peering supplies private IP routing between the VPCs; it does not provide transitive routing.

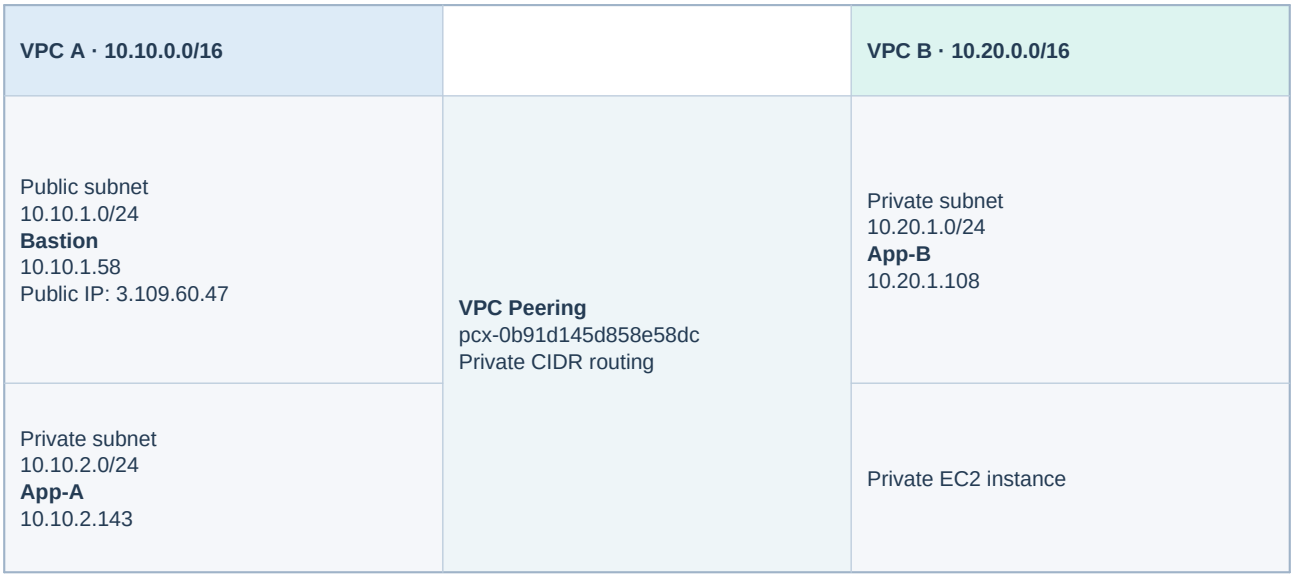


Figure 1. Logical network layout.

Resource	Configuration / observed value
VPC A	ewb-vpc-a · 10.10.0.0/16
VPC B	ewb-vpc-b · 10.20.0.0/16
Peering	pcx-0b91d145d858e58dc · Active
Bastion	ewb-web-server · 10.10.1.58 · 3.109.60.47
App-A	ewb-app-a · 10.10.2.143 · private
App-B	ewb-app-b · 10.20.1.108 · private

3. Routing and Peering Configuration

The peering connection was active. The following routes were confirmed during the lab.

Route table	Destination	Target
VPC A public	10.10.0.0/16	local
VPC A public	0.0.0.0/0	Internet Gateway (ewb-igw-a)
VPC A public	10.20.0.0/16	VPC Peering connection
VPC A private	10.10.0.0/16	local
VPC A private	10.20.0.0/16	VPC Peering connection
VPC B private	10.20.0.0/16	local
VPC B private	10.10.0.0/16	VPC Peering connection

Each subnet must be associated with the intended route table. App-A and App-B should remain private and should not have public IPv4 addresses.

4. Security Group Design

Security group	Purpose	Rule intent
ewb-web-sg	Bastion	Inbound SSH (TCP 22) from administrator's current public IP /32; avoid 0.0.0.0/0.
ewb-app-a-sg	App-A	Allow required private connectivity; SSH from bastion private IP /32 if needed.
ewb-app-b-sg	App-B	Allow ICMP Echo Request from App-A 10.10.2.143/32 for ping; allow SSH only from intended bastion / endpoint path.

A temporary ICMP rule change was used for the controlled failure test and restored afterward. Review the live AWS console before cleanup.

5. Validation Results

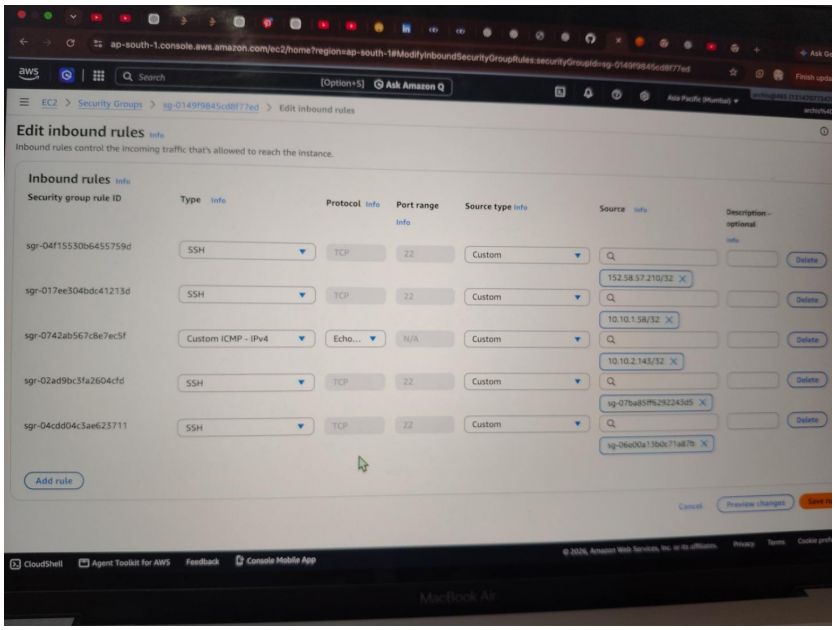
Test	Observed result	Status
App-A → App-B ping	4 packets received; 0% packet loss	PASS
App-B → App-A ping	4 packets received; 0% packet loss	PASS
SSH to App-B via bastion	SSH reached App-B; hostname / private IP checked	PASS
Remove App-B ICMP rule	Ping failed with 100% packet loss	EXPECTED FAILURE
Restore App-B ICMP rule	Ping succeeded again; 4 received, 0% loss	RECOVERED

Interpretation

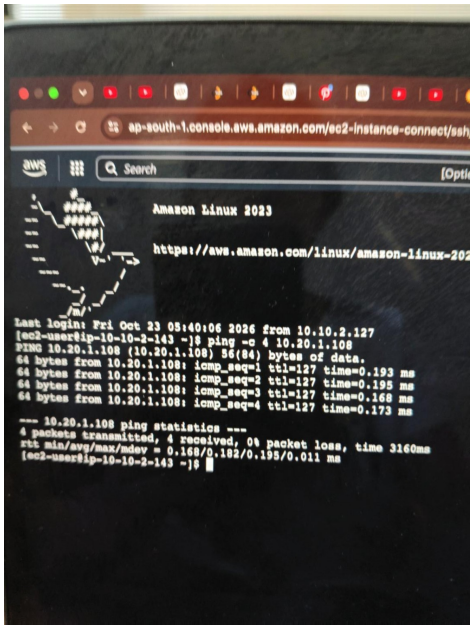
Successful pings in both directions show that peer routes and ICMP security rules were aligned for the tested endpoints. The controlled failure and recovery show that security group permissions affected reachability.

6. Screenshot Evidence

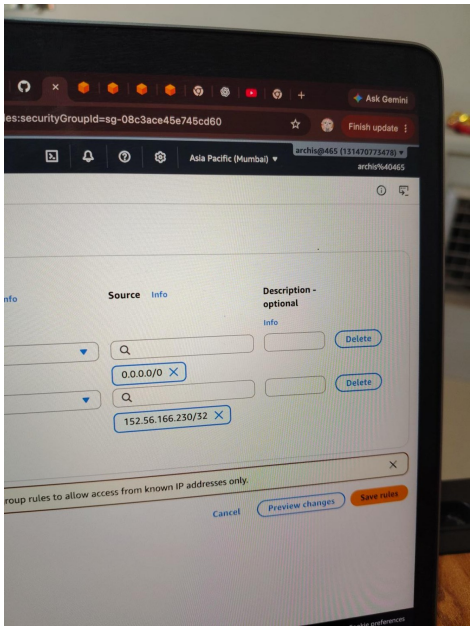
Screenshots are included from the available lab session. Some are photographs of the screen, so text may be less crisp than a direct screenshot. Replace or supplement with clear AWS-console or terminal captures if required.



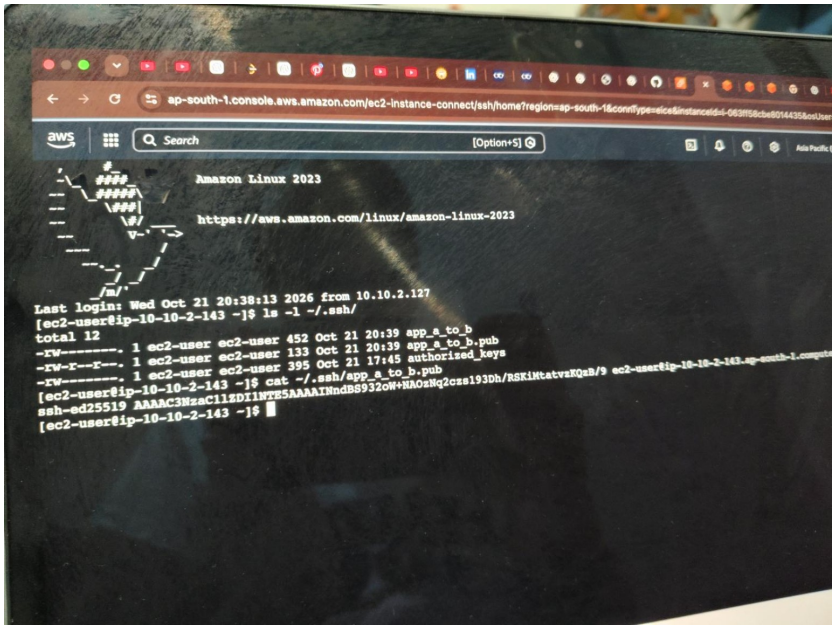
AWS console: EC2 instances / resource overview



Terminal: connectivity validation



AWS console: route or subnet configuration



Terminal: private instance access

Evidence checklist

- ■ VPC CIDR ranges and subnet CIDRs
- ■ Active VPC Peering connection
- ■ Public and private route tables
- ■ Both ping directions
- ■ SSH to App-B through bastion
- ■ ICMP failure and recovery

7. AWS Resource Cleanup

After completing the project, I reviewed the AWS resources used in the lab and checked the cleanup items below to help prevent unnecessary resource usage and charges.

Cleanup activities

• I reviewed the EC2 instances created for the VPC Peering project.

• I checked the VPC Peering connection and the route-table entries associated with both VPCs.

• I reviewed the security groups and networking resources used during testing.

• I checked for unused resources, including any EC2 Instance Connect Endpoint or Elastic IP address created for the lab.

• I reviewed the AWS account for remaining lab resources that could incur charges.

Note: This section describes the cleanup review. It does not claim that resources were deleted unless that action was completed and verified in the AWS console.

8. Learning Outcomes

This lab reinforced how VPC CIDR planning, route-table entries, peering state, and security group rules work together. VPC Peering creates a private network path, but communication succeeds only when both sides have correct routes and the destination security group permits the required traffic. The controlled failure experiment helped distinguish routing connectivity from security-group authorization.